

Overview of IEC 60601-1 Standards and References

Safety and Essential Performance of Medical Electrical Equipment

What is IEC 60601?

The **IEC 60601** series is an internationally recognized standard for the safety and essential performance of medical electrical equipment.

It sets requirements for the design, construction, and testing of electrical medical devices to ensure their safety for both patients and healthcare providers. The series includes several important sub-standards, with **IEC 60601-1** as the base standard, **IEC 60601-1-XX** as the collateral standards, and **IEC 60601-2-XX** as the particular standards.

- **IEC 60601-1 (Base Standard)**: This standard outlines general safety and performance requirements for all medical electrical equipment, ensuring that devices function correctly and are safe for patient use.
- **IEC 60601-1-XX (Collateral Standards)**: These are additional standards that address specific aspects of medical electrical equipment, such as electromagnetic compatibility, usability, and alarms. They complement the base standard and are essential for specific use cases in medical environments.
- **IEC 60601-2-XX (Particular Standards)**: These standards apply to specific types of medical devices, ensuring that unique requirements and safety concerns related to specific equipment (e.g., defibrillators, infusion pumps) are addressed.

IEC 60601-1:

IEC 60601-1 is the fundamental standard for the safety and essential performance of medical electrical equipment. It sets out the basic requirements for electrical medical devices to ensure that they do not pose any risk to patients, users, or the environment during normal use or under abnormal conditions. The standard covers general safety requirements for electrical equipment, including mechanical hazards, electrical hazards, and requirements for the overall functionality of the device.

GENERAL STANDARD		
IEC 60601-1	IEC 60601-1:2005 + AMD1:2012 - 5 - + AMD2:2020	IEC 60601-1 Edition 3.2 2020-08 "Medical electrical equipment – Part 1: General requirements for basic safety and essential performance"

Collateral Standards (IEC 60601-1-XX):

The **IEC 60601-1-XX** standards are referred to as collateral standards. These standards provide additional requirements for medical electrical equipment that are not covered in the base IEC 60601-1 standard. Each collateral standard focuses on a specific aspect of the equipment's safety or performance. These standards are designed to be applied in conjunction with IEC 60601-1 and provide a more detailed and specific set of requirements that address issues such as electromagnetic interference, usability, and alarm systems.

COLLATERAL STANDARDS		
IEC 60601-1-2	IEC 60601-1-2:2014/AMD1:2020	IEC 60601-1-2 Edition 4.0 2020-09 AMENDMENT 1 "Medical electrical equipment – Part 1-2: General requirements for basic safety and essential performance – Collateral Standard: Electromagnetic disturbances – Requirements and tests"
IEC 60601-1-3	IEC 60601-1-3:2008+AMD1:2013 +AMD2:2021	IEC 60601-1-3 Edition 2.2 2021-01 "Medical electrical equipment – Part 1-3: General requirements for basic safety and essential performance – Collateral Standard: Radiation protection in diagnostic X-ray equipment"
IEC 60601-1-6	IEC 60601-1-6:2010+AMD1:2013 +AMD2:2020	IEC 60601-1-6 Edition 3.2 2020-07 "Medical electrical equipment – Part 1-6: General requirements for basic safety and essential performance – Collateral standard: Usability"
IEC 60601-1-8	IEC 60601-1-8:2006+AMD1:2012 +AMD2:2020	IEC 60601-1-8 Edition 2.2 2020-07 "Medical electrical equipment – Part 1-8: General requirements for basic safety and essential performance – Collateral standard: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems"
IEC 60601-1-9	IEC 60601-1-9:2001+AMD1:2013 +AMD2:2020	IEC 60601-1-9 Edition 1.2 2020-07 "Medical electrical equipment – Part 1-9: General requirements for basic safety and essential performance – Collateral Standard: Requirements for environmentally conscious design"
IEC 60601-1-10	IEC 60601-1-10:2007 +AMD1:2013 +AMD2:2020	IEC 60601-1-10 Edition 1.2 2020-07 "Medical electrical equipment – Part 1-10: General requirements for basic safety and essential performance – Collateral Standard: Requirements for the development of physiologic closed-loop controllers"
IEC 60601-1-11	IEC 60601-1-11:2015+AMD1:2020	IEC 60601-1-11 Edition 2.1 2020-07 "Medical electrical equipment – Part 1-11: General requirements for basic safety and essential performance – Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment"
IEC 60601-1-12	IEC 60601-1-12:2014+AMD1:2020	IEC 60601-1-12 Edition 1.1 2020-07 "Medical electrical equipment – Part 1-12: General requirements for basic safety and essential performance – Collateral Standard: Requirements for medical electrical equipment and medical electrical systems intended for use in the emergency medical services environment"
IEC 62366-1	IEC 62366-1:2015/AMD1:2020	IEC 62366-1 Edition 1.0 2020-06 AMENDMENT 1 "Medical devices – Part 1: Application of usability engineering to medical devices"

Particular Standards (IEC 60601-2-XX)

The **IEC 60601-2-XX** standards are known as particular standards. These standards apply to specific types of medical electrical equipment and define the safety and performance requirements unique to those devices. They build on the base and collateral standards but address the specialized needs of individual device types, such as pacemakers, defibrillators, infusion pumps, and more. The goal is to ensure that medical devices operate safely and effectively in their specific intended medical contexts.

PARTICULAR STANDARDS		
IEC 60601-2-1	IEC 60601-2-1 Edition 4.0 (2020-10)	IEC 60601-2-1 Edition 4.0 (2020-10) "Medical electrical equipment – Part 2-1: Particular requirements for the basic safety and essential performance of electron accelerators in the range 1 MeV to 50 MeV"
IEC 60601-2-2	IEC 60601-2-2:2017+AMD1:2023	IEC 60601-2-2 Edition 6.1 2023-02 "Medical electrical equipment – Part 2-2: Particular requirements for the basic safety and essential performance of high frequency surgical equipment and high frequency surgical accessories"
IEC 60601-2-3	IEC 60601-2-3:2012+AMD1:2016 +AMD2:2022	IEC 60601-2-3:2012+AMD1:2016 +AMD2:2022 "Medical electrical equipment – Part 2-3: Particular requirements for the basic safety and essential performance of short-wave therapy equipment "
IEC 60601-2-4	IEC 60601-2-6:2012+AMD1:2016 +AMD2:2022	IEC 60601-2-6 Edition 2.2 2022-09 "Medical electrical equipment – Part 2-6: Particular requirements for the basic safety and essential performance of microwave therapy equipment"
IEC 60601-2-5	IEC 60601-2-5 Edition 3.0 (2009-07)	IEC 60601-2-5 Edition 3.0 (2009-07) "Medical electrical equipment – Part 2-5: Particular requirements for the basic safety and essential performance of ultrasonic physiotherapy equipment"
IEC 60601-2-6	IEC 60601-2-6:2012+AMD1:2016 +AMD2:2022	IEC 60601-2-6 Edition 2.2 2022-09 "Medical electrical equipment – Part 2-6: Particular requirements for the basic safety and essential performance of microwave therapy equipment"
IEC 60601-2-10	IEC 60601-2-10:2012 +AMD1:2016 +AMD2:2023	IEC 60601-2-10 Edition 2.2 2023-01 "Medical electrical equipment – Part 2-10: Particular requirements for the basic safety and essential performance of nerve and muscle stimulators"
ISO 80601-2-12	ISO 80601-2-12 Third edition (2023-11)	ISO 80601-2-12 Third edition (2023-11) "Medical electrical equipment – Part 2-12: Particular requirements for basic safety and essential performance of critical care ventilators"
ISO 80601-2-13	ISO 80601-2-13:2011/Amd.1:2015	ISO 80601-2-13:2011/Amd.1:2015 "Medical electrical equipment - Part 2-13: "Particular requirements for basic safety and essential performance of an anaesthetic workstation, AMD 1"
IEC 60601-2-16	IEC 60601-2-16:2018	IEC 60601-2-16 Edition 5.0 2018-04 "Medical electrical equipment – Part 2-16: Particular requirements for the basic safety and essential performance of haemodialysis, haemodiafiltration and haemofiltration equipment"
IEC 60601-2-18	60601-2-18:2009	IEC 60601-2-18 Edition 3.0 2009-08 "Medical electrical equipment – Part 2-18: Particular requirements for the basic safety and essential performance of endoscopic equipment"
IEC 60601-2-19	IEC 60601-2-19:2020 +AMD1:2023	IEC 60601-2-19:2020 +AMD1:2023 "Medical electrical equipment – Part 2-19: Particular requirements for the basic safety and essential performance of infant incubators"
IEC 60601-2-20	IEC 60601-2-20:2020	IEC 60601-2-20 Edition 3.0 2020-09 "Medical electrical equipment – Part 2-20: Particular requirements for the basic safety and essential performance of infant transport incubators"
IEC 60601-2-21	IEC 60601-2-21:2020 +AMD1:2023	IEC 60601-2-21:2020 +AMD1:2023 "Medical electrical equipment – Part 2-21: Particular requirements for the basic safety and essential performance of infant radiant warmers"
IEC 60601-2-22	IEC 60601-2-22:2019	IEC 60601-2-22 Edition 4.0 2019-11 "Medical electrical equipment – Part 2-22: Particular requirements for basic safety and essential performance of surgical, cosmetic, therapeutic and diagnostic laser equipment"
IEC 60601-2-24	IEC 60601-2-24:2012	IEC 60601-2-24 Edition 2.0 2012-10 "Medical electrical equipment – Part 2-24: Particular requirements for the basic safety and essential performance of infusion pumps and controllers"
IEC 60601-2-25	IEC 60601-2-25:2011	IEC 60601-2-25 Edition 2.0 2011-10 "Medical electrical equipment – Part 2-25: Particular requirements for the basic safety and essential performance of electrocardiographs"
IEC 80601-2-26	IEC 80601-2-26 Edition 1.0 (2019-05)	IEC 80601-2-26 Edition 1.0 (2019-05) "Medical electrical equipment – Part 2-26: Particular requirements for the basic safety and essential performance of electroencephalographs"
IEC 60601-2-27	IEC 60601-2-27:2011	IEC 60601-2-27 Edition 3.0 2011-03 "Medical electrical equipment – Part 2-27: Particular requirements for the basic safety and essential performance of electrocardiographic monitoring equipment"
IEC 60601-2-29	IEC 60601-2-29 Edition 3.0 (2008-06)	IEC 60601-2-29 Edition 3.0 (2008-06) "Medical electrical equipment – Part 2-29: Particular requirements for the basic safety and essential performance of radiotherapy simulators"
IEC 80601-2-30	IEC 80601-2-30 Edition 2.0 (2018-03)	IEC 80601-2-30 Edition 2.0 (2018-03) "Medical electrical equipment – Part 2-30: Particular requirements for the basic safety and essential performance of automated non-invasive sphygmomanometers"
IEC 60601-2-33	IEC 60601-2-33:2022/ISH1:2023	IEC 60601-2-33 Edition 4.0 2022-08 "Medical electrical equipment – Part 2-33: Particular requirements for the basic safety and essential performance of magnetic resonance equipment for medical diagnosis"
IEC 60601-2-34	IEC 60601-2-34 Edition 4.0 (2024-10)	IEC 60601-2-34 Edition 4.0 (2024-10) "Medical electrical equipment – Part 2-34: Particular requirements for the basic safety and essential performance of invasive blood pressure monitoring equipment"
IEC 60601-2-35	IEC 60601-2-35:2020 +AMD1:2023	IEC 60601-2-35:2020 +AMD1:2023 "Medical electrical equipment – Part 2-35: Particular requirements for the basic safety and essential performance of heating devices using blankets, pads or mattresses and intended for heating in medical use
IEC 60601-2-36	IEC 60601-2-36:2014	IEC 60601-2-36 Edition 2.0 2014-04 "Medical electrical equipment – Part 2-36: Particular requirements for the basic safety and essential performance of equipment for extracorporeally induced lithotripsy"
IEC 60601-2-37	IEC 60601-2-37 Edition 3.0 (2024-07)	IEC 60601-2-37 Edition 3.0 (2024-07) "Medical electrical equipment – Part 2-37: Particular requirements for the basic safety and essential performance of ultrasonic medical diagnostic and monitoring equipment"
IEC 60601-2-39	IEC 60601-2-39 Edition 3.0 (2018-04)	IEC 60601-2-39 Edition 3.0 (2018-04) "Medical electrical equipment – Part 2-39: Particular requirements for basic safety and essential performance of peritoneal dialysis equipment"
IEC 60601-2-40	IEC 60601-2-40:2016	IEC 60601-2-40 Edition 2.0 2016-08 "Medical electrical equipment – Part 2-40: Particular requirements for the basic safety and essential performance of electromyographs and evoked response equipment"
IEC 60601-2-41	IEC 60601-2-41 Edition 2.0 (2013-10)	IEC 60601-2-41 Edition 2.0 (2013-10) "Medical electrical equipment – Part 2-41: Particular requirements for the basic safety and essential performance of surgical luminaires and luminaires for diagnosis Amendment 1"
IEC 60601-2-43	IEC 60601-2-43:2022	IEC 60601-2-43 Edition 3.0 2022-12 "Medical electrical equipment – Part 2-43: Particular requirements for the basic safety and essential performance of X-ray equipment for interventional procedure"
IEC 60601-2-44	IEC 60601-2-44:2009 +AMD1:2012 +AMD2:2016	IEC 60601-2-44:2009 +AMD1:2012 +AMD2:2016 "Medical electrical equipment – Part 2-44: Particular requirements for the basic safety and essential performance of X-ray equipment for computed tomography"
IEC 60601-2-45	IEC 60601-2-45:2011 +AMD1:2015 +AMD2:2022	"Medical electrical equipment – Part 2-45: Particular requirements for the basic safety and essential performance of mammographic X-ray equipment and mammographic stereotactic devices"
IEC 60601-2-46	IEC 60601-2-46 Edition 4.0 (2023-05)	IEC 60601-2-46 Edition 4.0 (2023-05) "Medical electrical equipment – Part 2-46: Particular requirements for the basic safety and essential performance of operating tables"
IEC 60601-2-47	IEC 60601-2-47:2012	IEC 60601-2-47 Edition 2.0 2012-02 "Medical electrical equipment – Part 2-47: Particular requirements for the basic safety and essential performance of ambulatory electrocardiographic systems"
IEC 80601-2-49	IEC 80601-2-49:2018	IEC 80601-2-49 Edition 1.0 2018-03 "Medical electrical equipment – Part 2-49: Particular requirements for the basic safety and essential performance of multifunction patient monitors"
IEC 60601-2-50	IEC 60601-2-50:2020 +AMD1:2023	IEC 60601-2-50:2020 +AMD1:2023 "Medical electrical equipment – Part 2-50: Particular requirements for the basic safety and essential performance of infant phototherapy equipment"
IEC 60601-2-52	IEC 60601-2-52:2009 +AMD1:2015	IEC 60601-2-52 Edition 1.1 2015-03 "Medical electrical equipment – Part 2-52: Particular requirements for the basic safety and essential performance of medical beds"
IEC 60601-2-54	IEC 60601-2-54:2022	IEC 60601-2-54 Edition 2.0 2022-09 "Medical electrical equipment – Part 2-54: Particular requirements for the basic safety and essential performance of X-ray equipment for radiography and radioscopy"
ISO 80601-2-55	ISO 80601-2-55 First edition (2011-12-15)	ISO 80601-2-55 First edition (2011-12-15) "Medical electrical equipment - Part 2-55: Particular requirements for the basic safety and essential performance of respiratory gas monitors "
ISO 80601-2-56	ISO 80601-2-56:2017+A1:2020	ISO 80601-2-56:2017 +A1:2020 (May 2020) "Medical electrical equipment - Part 2-56: Particular requirements for basic safety and essential performance of clinical thermometers for body temperature measurement"
IEC 60601-2-57	IEC 60601-2-57:2023	IEC 60601-2-57 Edition 2.0 2023-07 "Medical electrical equipment – Part 2-57: Particular requirements for the basic safety and essential performance of non-laser light source equipment intended for therapeutic, diagnostic, monitoring, cosmetic and aesthetic use
IEC 80601-2-58	IEC 80601-2-58:2014+AMD1:2016	IEC 80601-2-58 Edition 2.1 2016-10 "Medical electrical equipment – Part 2-58: Particular requirements for the basic safety and essential performance of lens removal devices and vitrectomy devices for ophthalmic surgery
IEC 80601-2-59	IEC 80601-2-59:2017 +AMD1:2023	IEC 80601-2-59 Edition 2.1 2023-01 "Medical electrical equipment – Part 2-59: Particular requirements for the basic safety and essential performance of screening thermographs for human febrile temperature screening"
IEC 80601-2-60	IEC 80601-2-60:2019	IEC 80601-2-60 Edition 2.0 2019-06 "Medical electrical equipment – Part 2-60: Particular requirements for the basic safety and essential performance of dental equipment"
ISO 80601-2-61	ISO 80601-2-61:2019	ISO 80601-2-61:2017, Corrected version 2018-02 "Medical electrical equipment - Part 2-61: Particular requirements for basic safety and essential performance of pulse oximeter equipment"
IEC 60601-2-62	IEC 60601-2-62 Edition 1.0 (2013-07)	IEC 60601-2-62 Edition 1.0 (2013-07) "Medical electrical equipment – Part 2-62: Particular requirements for the basic safety and essential performance of high intensity therapeutic ultrasound (HITU) equipment"
IEC 60601-2-63	IEC 60601-2-63:2012+A1:2017 +A2:2021	IEC 60601-2-63:2012 +A1:2017 +A2:2021 "Medical electrical equipment – Part 2-63: Particular requirements for the basic safety and essential performance of dental extra-oral X-ray equipment"
IEC 60601-2-64	IEC 60601-2-64 Edition 1.0 (2014-09)	IEC 60601-2-64 Edition 1.0 (2014-09) "Medical electrical equipment – Part 2-64: Particular requirements for the basic safety and essential performance of light ion beam medical electrical equipment"
IEC 60601-2-65	IEC 60601-2-65:2012 +AMD1:2017 +AMD2:2021	"Medical electrical equipment – Part 2-65: Particular requirements for the basic safety and essential performance of dental intra-oral X-ray equipment "
IEC 60601-2-66	IEC 60601-2-66:2019	IEC 60601-2-66 Edition 3.0 2019-10 "Medical electrical equipment – Part 2-66: Particular requirements for the basic safety and essential performance of hearing aids and hearing aid systems"
ISO 80601-2-67	ISO 80601-2-67:2020	ISO 80601-2-67:2020 "Medical electrical equipment - Part 2-67: Particular requirements for basic safety and essential performance of oxygen-conserving equipment"
IEC 60601-2-68	IEC 60601-2-68 Edition 1.0 (2014-09)	IEC 60601-2-68 Edition 1.0 (2014-09) "Medical electrical equipment – Part 2-68: Particular requirements for the basic safety and essential performance of X-ray-based image-guided radiotherapy equipment for use with electron accelerators, light ion beam therapy equipment and radionuclide beam therapy equipment"
ISO 80601-2-69	ISO 80601-2-69:2020	ISO 80601-2-69:2020 "Medical electrical equipment - Part 2-69: Particular requirements for the basic safety and essential performance of oxygen concentrator equipment"
ISO 80601-2-70	ISO 80601-2-70:2020	ISO 80601-2-70:2020 "Medical electrical equipment - Part 2-70: Particular requirements for the basic safety and essential performance of sleep apnoea breathing therapy equipment"
IEC 80601-2-71	IEC 80601-2-71 Edition 1.0 (2015-06)	IEC 80601-2-71 Edition 1.0 (2015-06) "Medical electrical equipment – Part 2-71: Particular requirements for the basic safety and essential performance of functional near-infrared spectroscopy (NIRS) equipment"
ISO 80601-2-72	ISO 80601-2-72 First edition (2015-09-01)	ISO 80601-2-72 First edition (2015-09-01) "Medical electrical equipment - Part 2-72: Particular requirements for basic safety and essential performance of home healthcare environment ventilators for ventilator-dependent patients"
ISO 80601-2-74	ISO 80601-2-74:2021	ISO 80601-2-74:2021 "Medical electrical equipment - Part 2-74: Particular requirements for basic safety and essential performance of respiratory humidifying equipment
IEC 60601-2-75	IEC 60601-2-75:2017 +AMD1:2023	IEC 60601-2-75 Edition 1.1 2023-01 "Medical electrical equipment – Part 2-75: Particular requirements for the basic safety and essential performance of photodynamic therapy and photodynamic diagnosis equipment"
IEC 60601-2-76	IEC 60601-2-76:2018 +AMD1:2023	IEC 60601-2-76 Edition 1.1 2023-08 "Medical electrical equipment – Part 2-76: Particular requirements for the basic safety and essential performance of low energy ionized gas haemostasis equipment"
IEC 80601-2-77	IEC 80601-2-77:2019	IEC 80601-2-77 Edition 1.0 2019-07 "Medical electrical equipment – Part 2-77: Particular requirements for the BASIC SAFETY and essential performance of ROBOTICALLY ASSISTED SURGICAL EQUIPMENT"
IEC 80601-2-78	IEC 80601-2-78 Edition 1.0 (2019-07)	IEC 80601-2-78 Edition 1.0 (2019-07) "Medical electrical equipment – Part 2-78: Particular requirements for basic safety and essential performance of medical robots for rehabilitation, assessment, compensation or alleviation"
ISO 80601-2-79	ISO 80601-2-79 First edition (2018-07)	ISO 80601-2-79 First edition (2018-07) "Medical electrical equipment - Part 2-79: Particular requirements for basic safety and essential performance of ventilatory support equipment for ventilatory impairment"
ISO 80601-2-80	ISO 80601-2-80 First edition (2018-07)	ISO 80601-2-80 First edition (2018-07) "Medical electrical equipment - Part 2-80: Particular requirements for basic safety and essential performance of ventilatory support equipment for ventilatory insufficiency"
IEC 60601-2-83	IEC 60601-2-83:2019 +AMD1:2022	IEC 60601-2-83 Edition 1.1 2022-12 "Medical electrical equipment – Part 2-83: Particular requirements for the basic safety and essential performance of home light therapy equipment"
ISO 80601-2-84	ISO 80601-2-84 First edition (2020-07)	ISO 80601-2-84 First edition (2020-07) "Medical electrical equipment - Part 2-84: Particular requirements for the basic safety and essential performance of ventilators for the emergency medical services environment"
ISO 80601-2-85	ISO 80601-2-85:2021 first edition (2021-03)	ISO 80601-2-85:2021 First edition (2021-03) "Medical electrical equipment - Part 2-85: Particular requirements for the basic safety and essential performance of cerebral tissue oximeter equipment"
ISO 80601-2-87	ISO 80601-2-87:2021 first edition (2021-04)	ISO 80601-2-87:2021 First Edition (2021-04) "Medical electrical equipment - Part 2-87: Particular requirements for basic safety and essential performance of high-frequency ventilators"
ISO 80601-2-90	ISO 80601-2-90:2021	ISO 80601-2-90:2021 "Medical electrical equipment - Part 2-90: Particular requirements for basic safety and essential performance of respiratory high-flow therapy equipment"
ISO 10079-1	ISO 10079-1: 2022	ISO 10079-1 (2022) "Medical suction equipment - Part 1: electrically powered suction equipment"
ISO 10079-4	ISO 10079-4:2021	ISO 10079-4 (2021) "Medical suction equipments - Part 4: General requirements"

Intertek Solutions for IEC 60601-1

As a [Nationally Recognized Testing Laboratory \(NRTL\) approved by OSHA](#), Intertek is an industry leader providing Testing, Certification and in-lab support to help you navigate the requirements of the IEC 60601 series of standards. With our team of experts globally, Intertek is positioned to help determine standard applicability and execution. Through standards testing, you can support your safety claims throughout North America, Europe and beyond. We offer end-to-end solutions from product development, to review of the Risk Management

File and supporting documentation, to comprehensive testing to the 60601/80601 series of standards, including an array of Collateral (IEC 60601-1-X) and Particular (IEC 60601-2-X) Standards. Intertek is your trusted partner throughout the testing process.

IEC 60601 Resources